# How to use IFDIFF

- Write code of rhs
- Setup time horizon, initial values, integrator options

```
tspan = [0 20];
x0    = [1 0];
p     = 5.437;
opts  = odeset('reltol', 1e-6);
```

- Single preparation call

```
dhandle = prepareDatahandleForIntegration( ...
             'rhs_f', 'solver', 'ode45', 'options', opts);
```

- Call integrator

```
sol = ode45(@(t,x) rhs_f(t,x,p), tspan, x0, opts);
sol = solveODE(dhandle, tspan, x0, p);
```

- Evaluate and use, plot, etc..

```
T = 0:0.1:20;
X = deval(sol, T);
plot(T,X);
```

```
function dx = rhs_f(t,x,p)
  dx = zeros(2,1);
  dx(1) = 0.01 * t.^2  +  x(2).^3;
  if x(1) < p(1)
     dx(2) = 0;
  else
     if x(1) < p(1) + 0.5
        dx(2) = 5;
     else
        dx(2) = 0;
     end
  end
end
```

File: rhs_f.m

- IFDIFF:
  - Unmodified rhs code
  - Single preparation call for the rhs file
  - Transparent usage of Matlab's sol structure (no further code modifications necessary)